

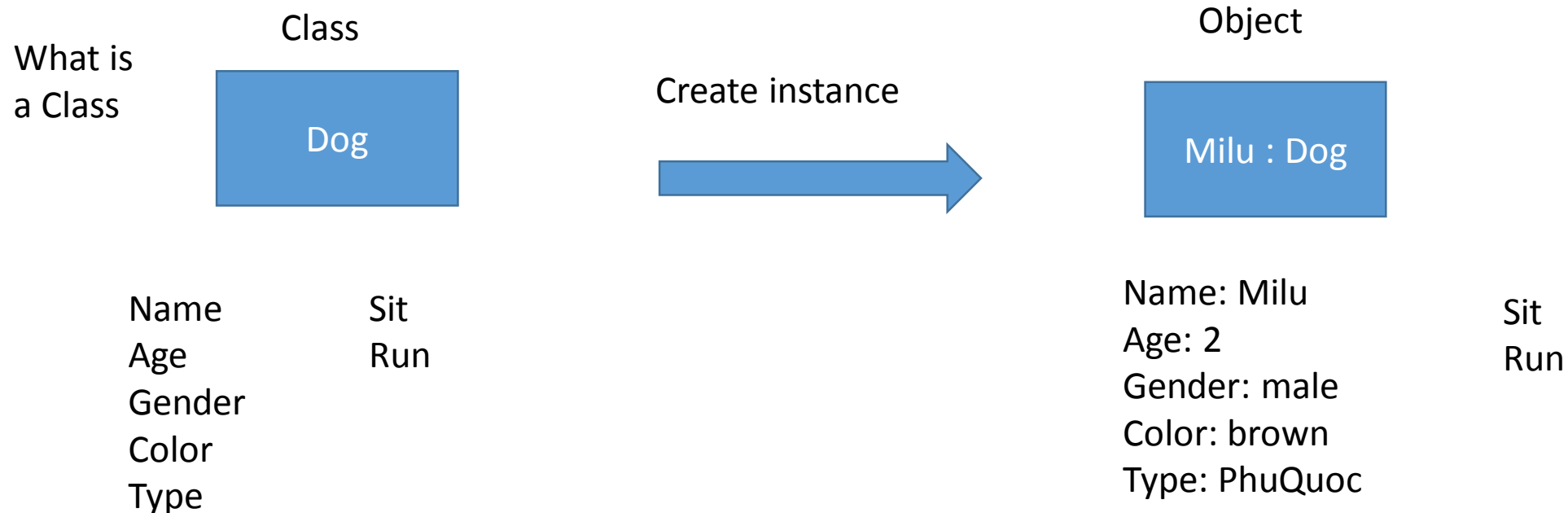
Class diagram guide

Notes for project #4

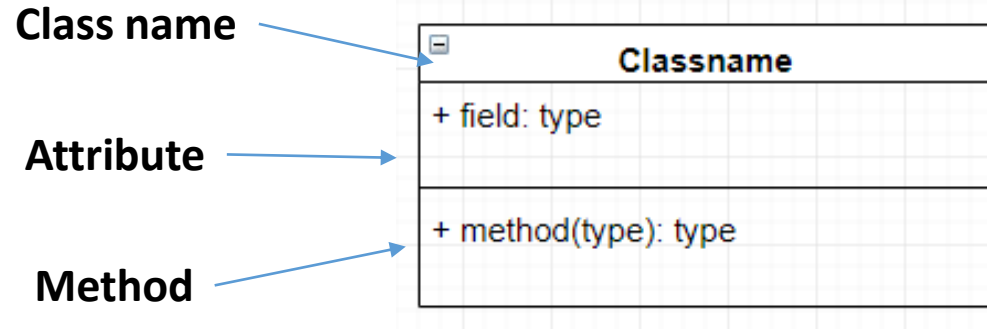
4/2019

What is a Class diagram

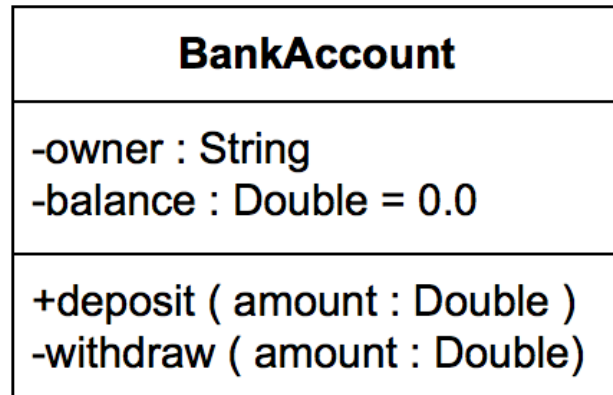
- Class diagrams, which show the object classes in the system and the associations between these classes.



Class



Example:
Bank account class



Visibility of Class Members

(attributes and methods)

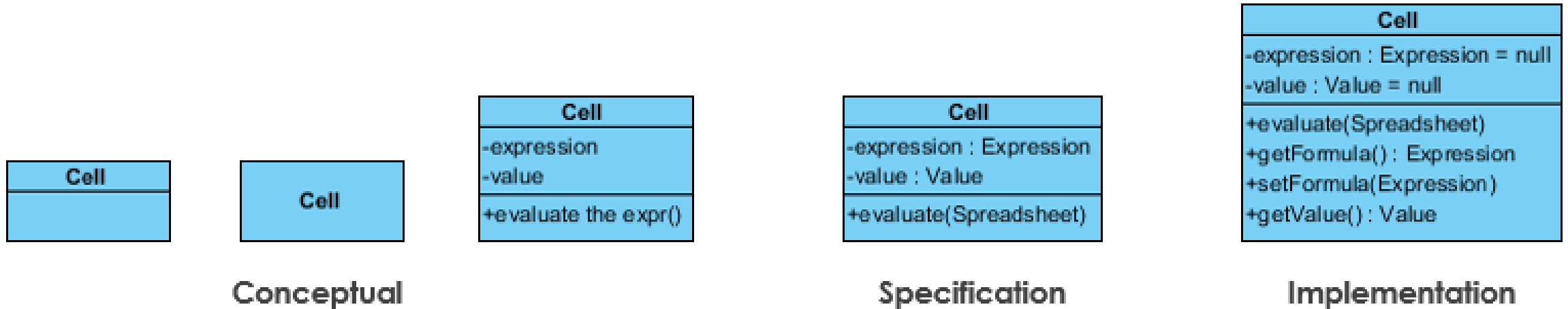
+ : public

- : private

: protected

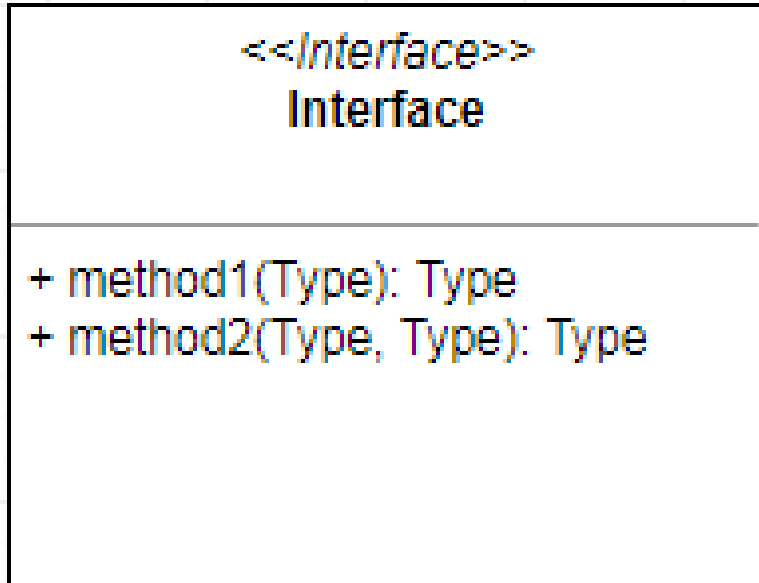
~ : package

Perspectives of Class Diagram



- **Conceptual:** represents the concepts in the domain
- **Specification:** focus is on the interfaces of Abstract Data Type (ADTs) in the software
- **Implementation:** describes how classes will implement their interfaces

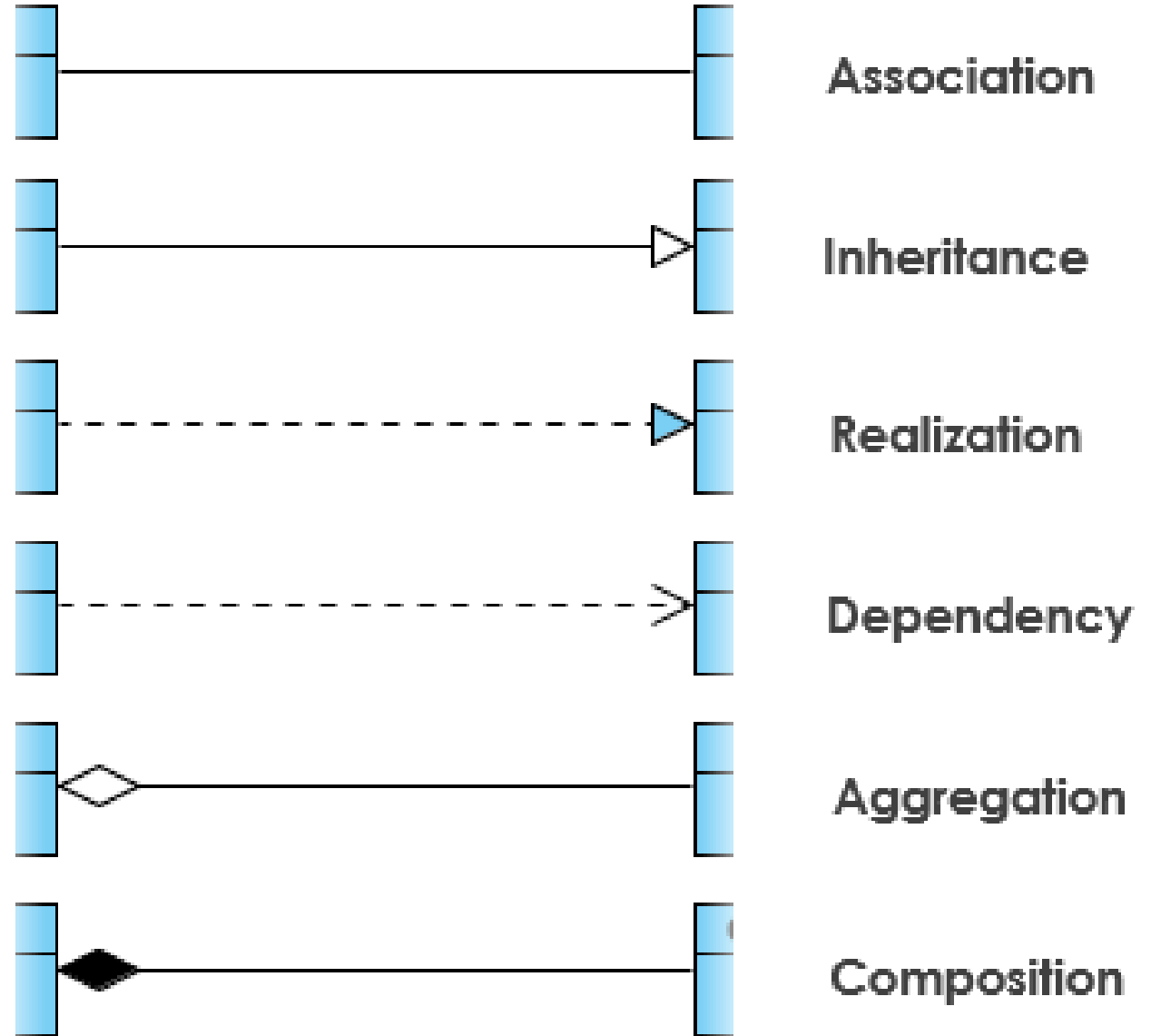
Interface



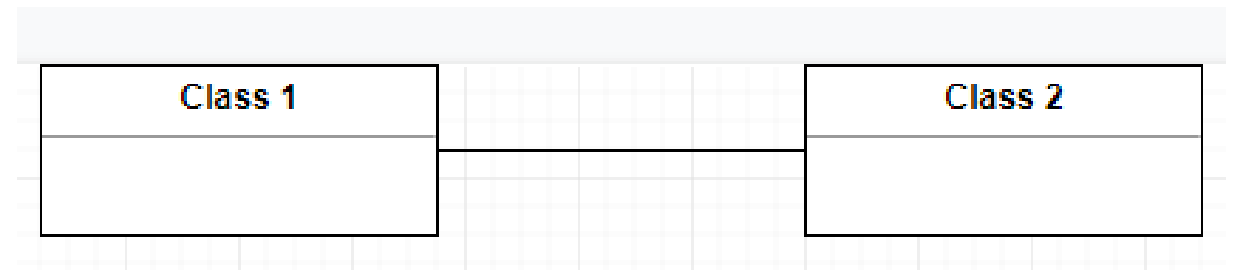
Abstract Class

Interface
vs
Abstract Class
???

Relationships between classes



Association

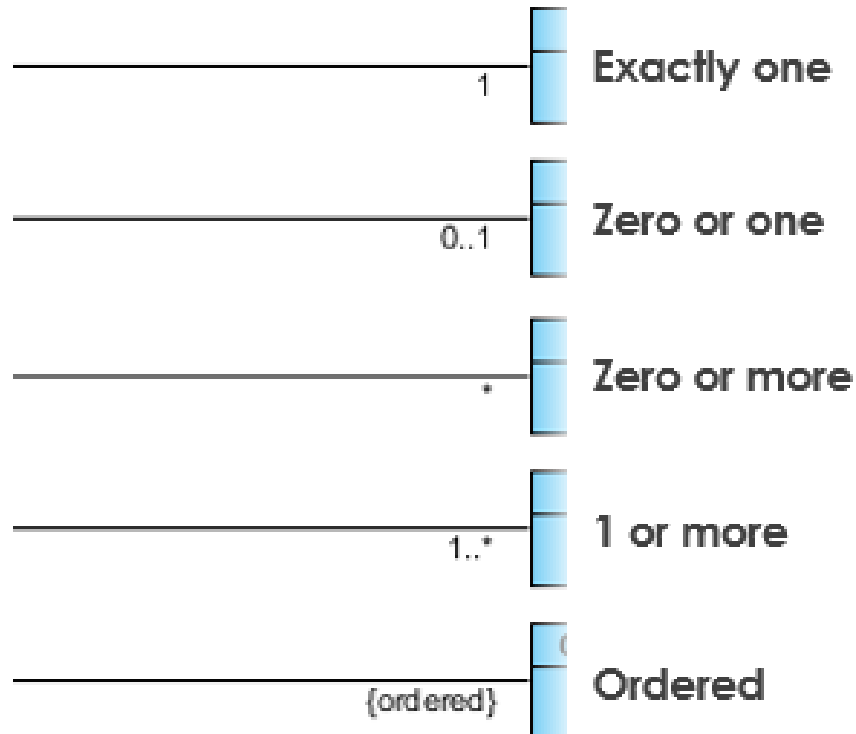


There is an association between Class1 and Class2

Simple Association

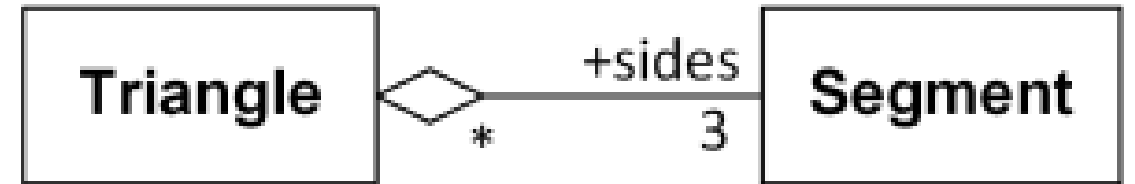
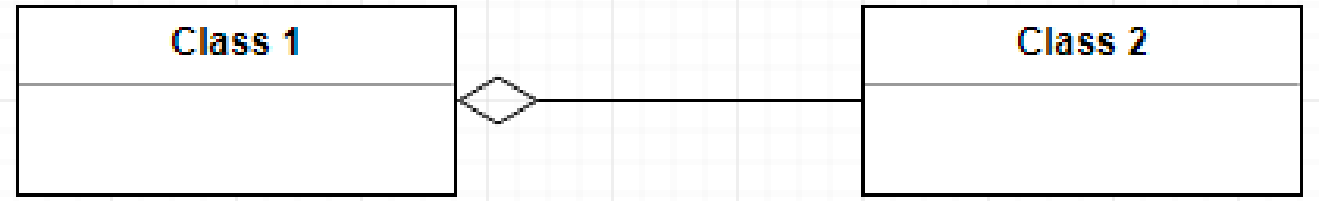
- A structural link between two peer classes.

Cardinality



Aggregation

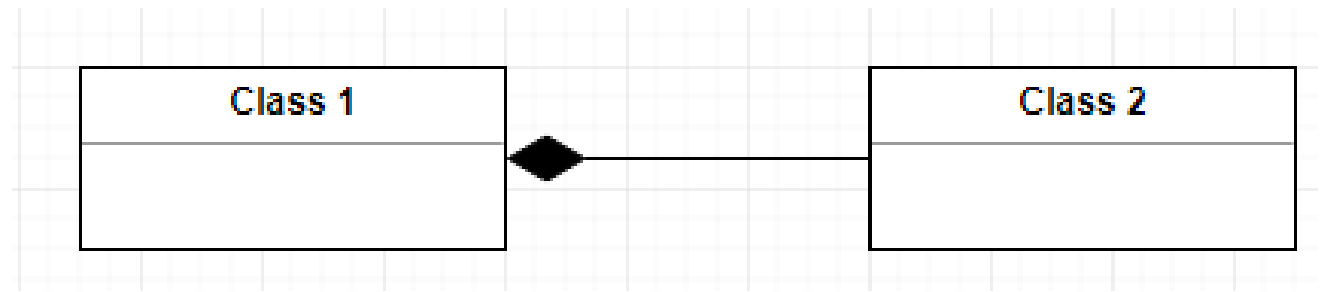
- A special type of association.
- It represents a "part of" relationship.
- Class2 is part of Class1.
- Many instances (denoted by the *) of Class2 can be associated with Class1.
- Objects of Class1 and Class2 have separate lifetimes.



*Triangle has 'sides' collection of three line Segments.
Each line Segment could be part of none, one, or several triangles.*

Composition

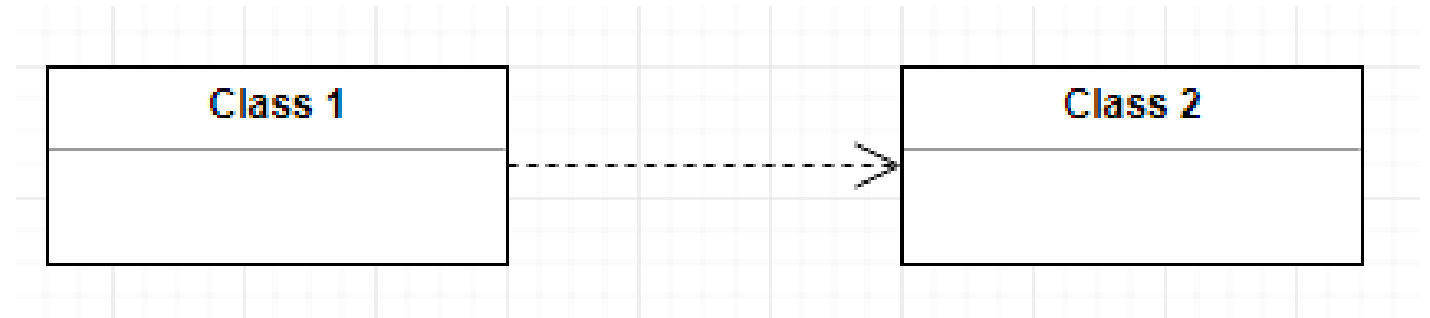
- A special type of aggregation where parts are destroyed when the whole is destroyed.
- Objects of Class2 live and die with Class1.
- Class2 cannot stand by itself.



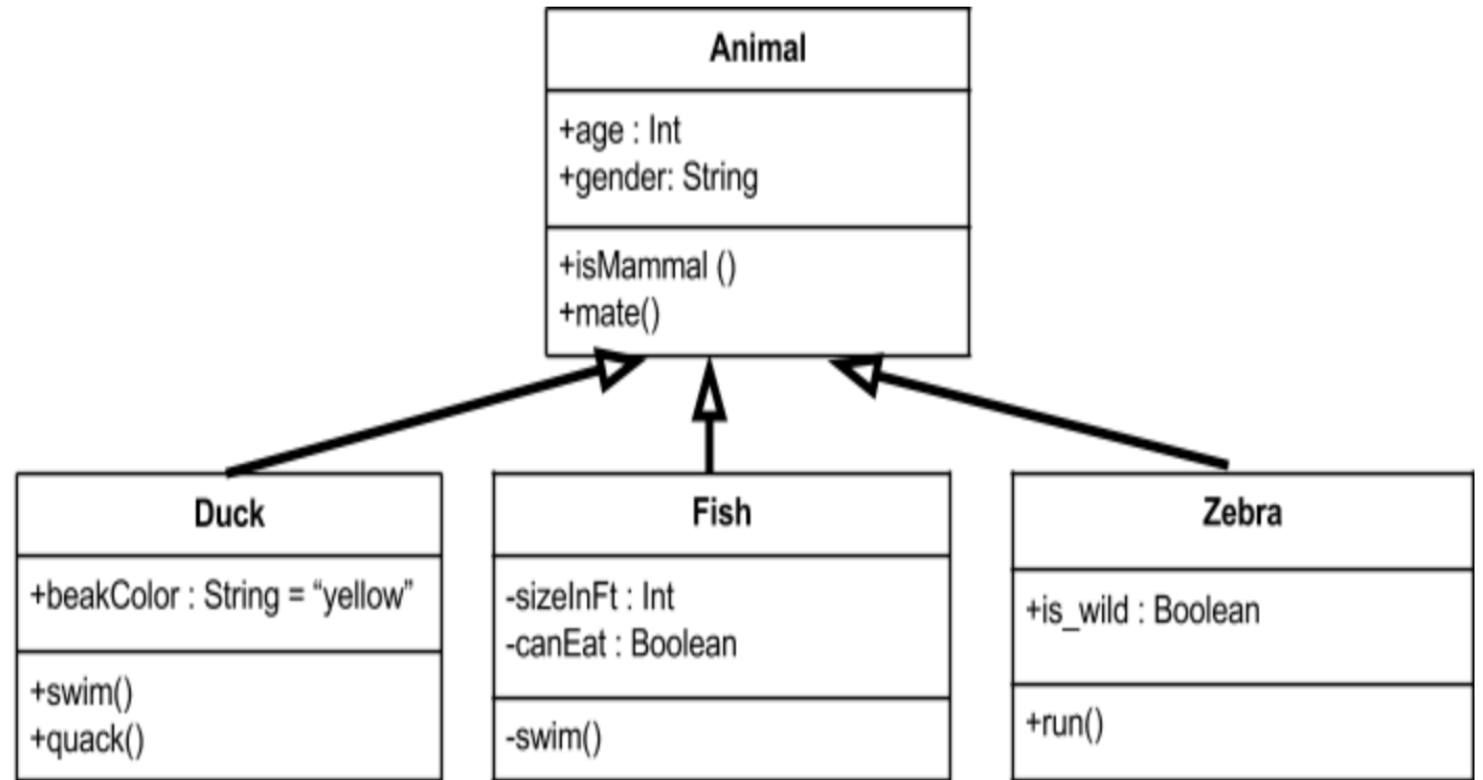
*Folder could contain many files, while each File has exactly one Folder parent.
If Folder is deleted, all contained Files are deleted as well.*

Dependency

- A special type of association.
- Exists between two classes if changes to the definition of one may cause changes to the other (but not the other way around).
- Class1 depends on Class2

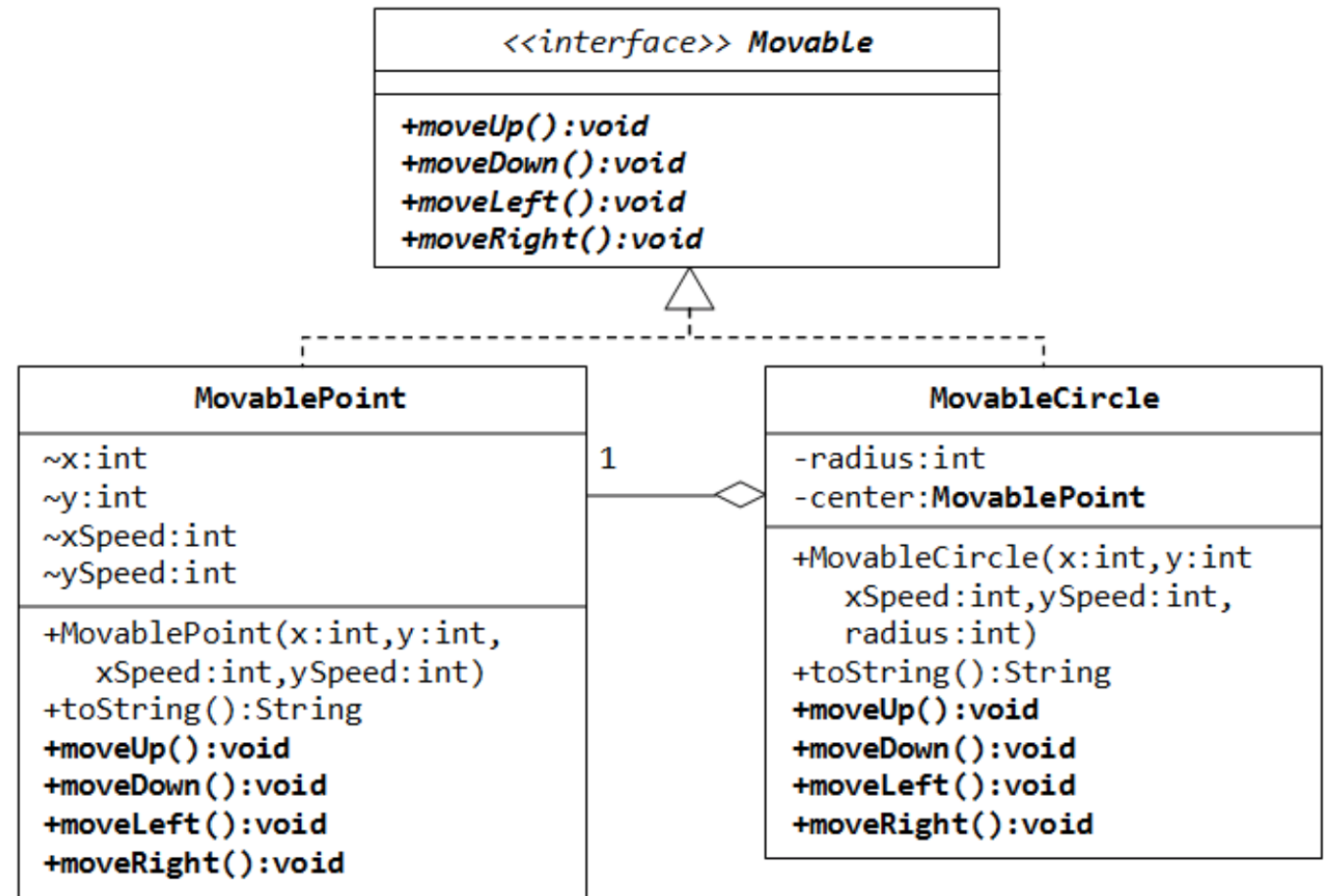


Inheritance (or Generalization)

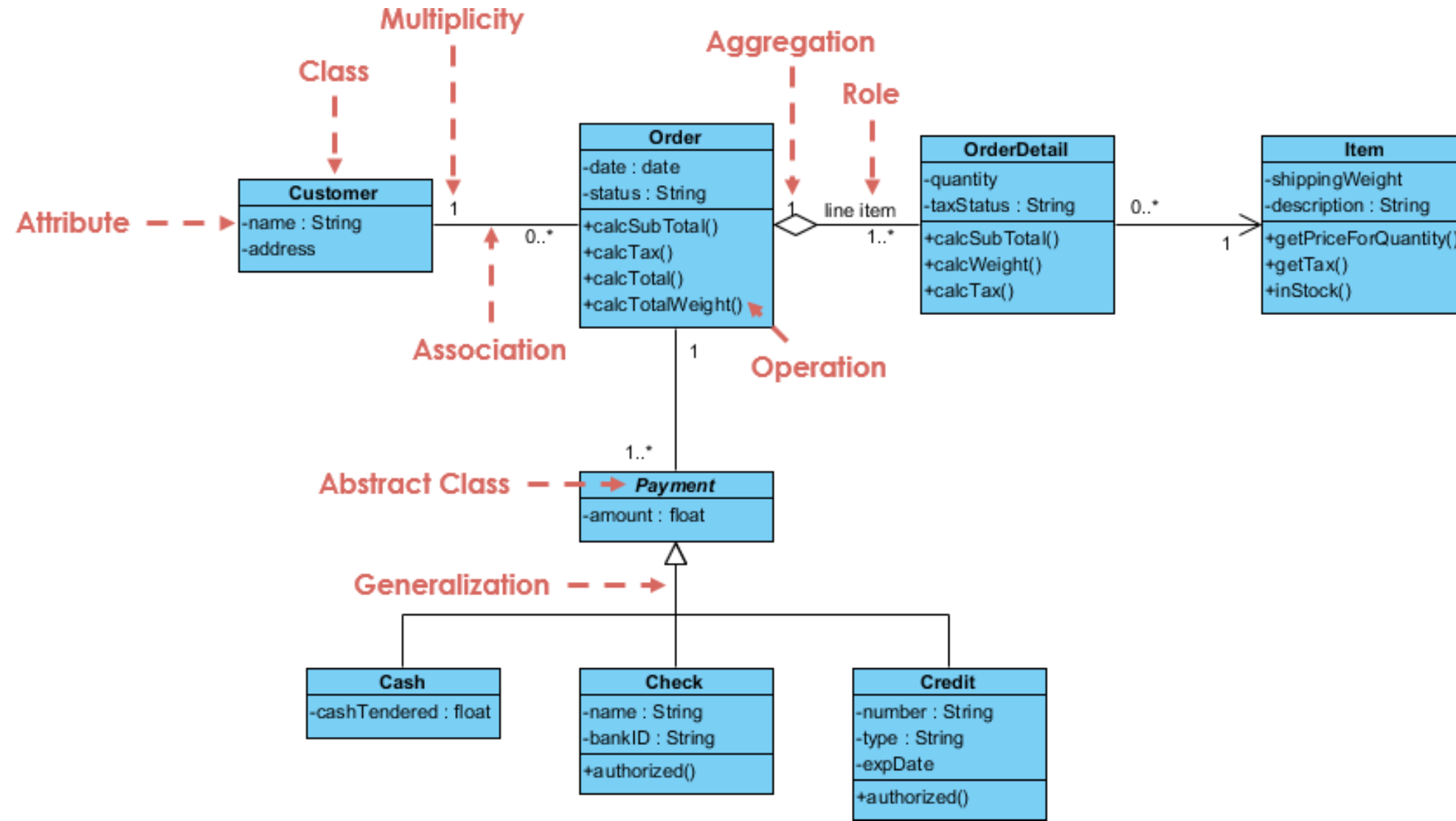


Realization/Implementation

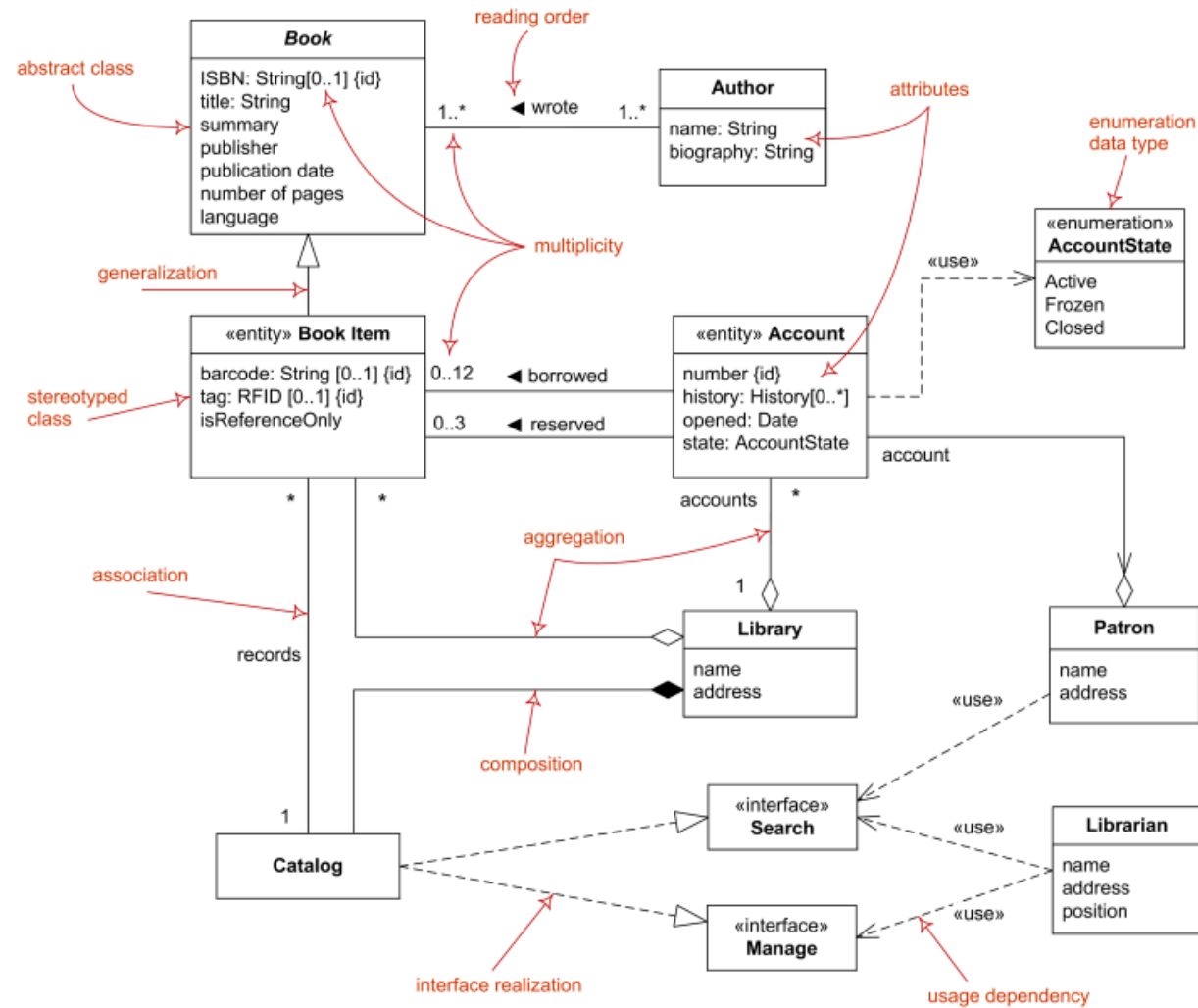
Realization is a specialized abstraction relationship between two sets of model elements, one representing a **specification** (the supplier) and the other represents an **implementation** of the latter (the client).



Class Diagram Example: Order System

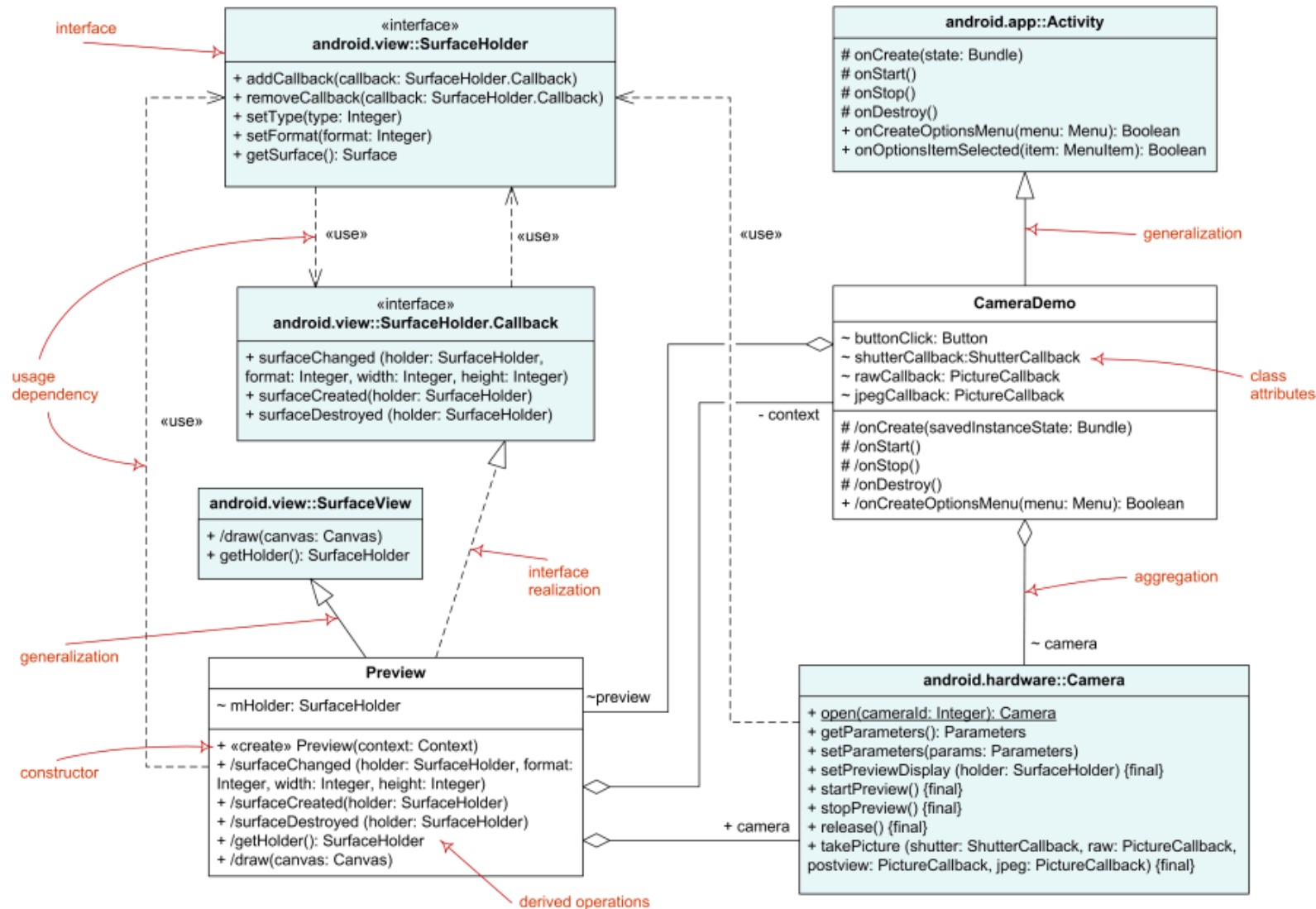


Domain Model Diagram



Domain diagram overview - classes, interfaces, associations, usage, realization, multiplicity.

Diagram of Implementation Classes



Elements of implementation class diagram - classes, interfaces, associations, usage, realization.

Design pattern

- Singleton pattern
- Proxy pattern
- Composite pattern
- Adapter pattern
- Facade pattern
-

<http://blog.genmymodel.com/discover-five-design-patterns-in-one-uml-class-diagram-example.html>

Project #4 submission

- **Module interface** cho các hàm dùng chung từ module này đến module khác
- **Class diagram** cho các chức năng cá nhân phụ trách
- **Methods description/design** cho tất cả các method của các class của cá nhân phụ trách
- Các **sequence diagram, activity diagram** bổ sung để giải thích cách thức tương tác, hoạt động (ở mức chi tiết)
- (Bonus) **Design pattern**: dùng design pattern trong thiết kế của mình

Note for Proj #4

- **Module interface**: dùng cách định nghĩa <<interface>> hoặc liệt kê danh sách các "method signature" cho từng module.
- **Class diagram**: cần phải có đủ entity/data class và control/business class và view/boundary class
- **Sequence diagram, activity diagram** : ở mức chi tiết (tên hàm)
- **Design pattern**: mô tả lại "chỗ nào"/"cái nào"/"cụm class nào" dùng và tại sao.

Project #4	Quá dưới mức yêu cầu (0% đ) No or almost no	Dưới mức yêu cầu (50% đ) Under expectation	Gần đạt yêu cầu (75% đ) Almost at the expectation	Đạt yêu cầu (100% đ) Expectation	Điểm Point
(Module) Interface design	Không có No 0	Thiếu, sơ sài Missing, simple 0.5	Đầy đủ, thiếu đặc tả tham số Missing signatures 0.75	Đầy đủ đến mức chi tiết Detail 1	1
Class diagram	Không có No 0	Thiếu hoặc chỉ gồm entity/data classes Missing OR only entity/data classes 1.5	Thiếu entity/data hoặc business/controller hoặc view class hoặc có sai sót ký hiệu Missing entity/data OR missing business/controller OR missing view classes OR mistakes 2	Có đầy đủ, đúng ký hiệu All classes, correct 3	3

Method description	Không có No 0	Thiếu nhiều và sơ sài Missing many methods and mistakes 2	Đủ, mô tả chưa rõ No clear description 3	Có đủ kể cả mô tả các method All (including method description) 4	4
Sequence diagram	Không có No 0	Không phải cấp design hoặc sai ký hiệu Many mistake or too general 0.5	Mức design nhưng còn thiếu Detail, missing some diagram 1	Có đủ giải thích và đúng Good 1.5	1.5
Activity/State diagram	Không có No 0	Không phải cấp design hoặc sai ký hiệu Many mistake or too general 0.25		Tốt Good 0.5	0.5
Design pattern	Không dùng No	Đơn giản (ví dụ MVC cho phần user-interface) Simple (ex. MVC) 0.5	Có dùng 1 design pattern phù hợp One good design pattern 0.75	Có dùng hơn 1 design pattern phù hợp More than one good design pattern 1	+1
Total					10